

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE								
	Winter Examination – 2022								
	Course: B. Tech.				Branch :Civil Enginerring				
	Semester : Fifth								
	Subject Code & Name: BTCVC 504 Concrete Technology								
	Max Marks: 60		Date: 08/02/23			Duration: 3 Hr.			
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.								
								(Level/CO)	Marks
Q. 1	Solve Any Two of the following.								12
A)	Define Hydration of cement. Explain wet Process of Cement manufacturing with flow chart.							CO1	6
B)	Define bulking of sand and explain in detail the procedure to find the bulking of sand with diagram..							CO1	6
C)	Calculate fineness modulus for the given data of fine aggregate. Total weight of C.A. = 1000 gm. State type of sand							CO1	6
	Sieve size (mm)	4.75	2.36	1.18	600μ	300μ	150μ	Pan	
	Weight retained (gm)	20	75	210	274	305	106	10	
Q.2	Solve Any Two of the following.								12
A)	Describe the procedure for determination of workability by slump cone test with diagram..							CO1	6
B)	Explain in detail Batching of concrete with its types.							CO1	6
C)	Write short note on i) Segregation ii) Bleeding							CO1	6
Q. 3	Solve Any Two of the following.								12
A)	Define Admixtures. Enlist its types and state purpose of adding the admixtures into the concrete. (any four)							CO2	6
B)	Enlist the different types of construction chemicals and state its applications.							CO2	6
C)	Write a short notes on Green Concrete and Bonding admixtures.							CO2	6
Q.4	Solve Any Two of the following.								12

A)	What is mean by Alkali-Aggregate Reaction (AAR) and write the points to control Alkali-Aggregate Reaction.	CO1	6
B)	Explain in detail shrinkage of concrete with its types and their control.	CO1	6
C)	State the factors contributing to cracks in concrete and its preventive measures.	CO1	6
Q. 5	Solve Any Two of the following.		12
A)	Explain the concrete mix design procedure by IS code method.	CO3	6
B)	What are the factors causing variations in the quality of concrete.	CO3	6
C)	Enlist Non Destructive tests for determination of strength of hardened concrete and explain any one test.	CO3	6
	*** End ***		

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